

HUMAN CONNECTOME · SESSION ANALYSIS REPORT

Session ID: session_1780455159141_cxu2a
Period: 2026-06-03T02:30:46.475Z → 2026-06-03T02:52:39.099Z
Wall-clock duration: 1312.6s | Ticks: 4,888 | Estimated tick interval: ~268.5ms
High-confidence thoughts (≥75%): 16

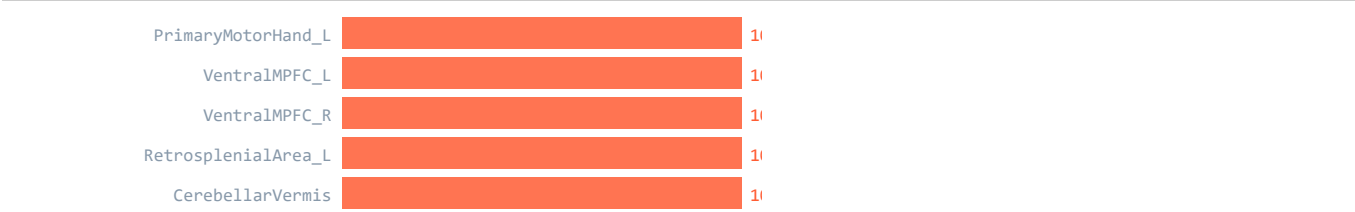
Scientific Validity Notice: Claim language used throughout: "emergent behavior in a brain-inspired embodied simulation." Shannon entropy is normalized as $H/\log(N_{\text{regions}})$; $\text{max}=1.0$. Session duration uses wall-clock time. Pearson r is correlation only – NOT transfer entropy. $r=\pm 1.000$ flagged as possible computation artifact. STDP = STDP-inspired population plasticity (Wilson-Cowan rate model). Platform biologically constrained by HCP-MMP1.0, Allen Brain Atlas, and Brainnetome Atlas data.

⚠ Saturation Warning: 5 region(s) averaged >90% activation: PrimaryMotorHand_L, VentralMPFC_L, VentralMPFC_R, RetrosplenialArea_L, CerebellarVermis. Verify homeostatic scaling before publishing emergence claims.

SESSION METRICS

DURATION 1312.6 seconds (wall-clock)	TICKS 4,888 simulation steps	SHANNON ENTROPY 0.839 $H/\log(N)$ · $\text{max}=1.0$	PLASTICITY INDEX 0.135 STDP-inspired · pop. rate
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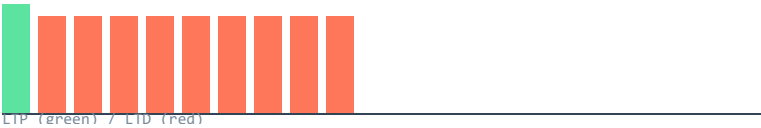
NEURAL ACTIVITY · TOP 15 REGIONS BY AVG ACTIVATION



#	REGION	AVG ACTIVATION	SATURATION FLAG
1	PrimaryMotorHand_L	100%	⚠ SATURATED – verify
2	VentralMPFC_L	100%	⚠ SATURATED – verify
3	VentralMPFC_R	100%	⚠ SATURATED – verify
4	RetrosplenialArea_L	100%	⚠ SATURATED – verify
5	CerebellarVermis	100%	⚠ SATURATED – verify

STDP-INSPIRED POPULATION PLASTICITY · WEIGHT CHANGES

Wilson-Cowan rate model. Eligibility-trace approximation. NOT literal spike-timing plasticity.



CONNECTION	DELTA	TYPE
Hippocampus->PrefrontalCortex	+0.7865	LTP
PrefrontalCortex->Thalamus	-0.7000	LTD
PrefrontalCortex->BasalGanglia	-0.7000	LTD
PrefrontalCortex->AnteriorCingulateCortex	-0.7000	LTD
SupplementaryMotorArea->MotorCortex	-0.7000	LTD

CONNECTION	DELTA	TYPE
CA1->Hippocampus	-0.7000	LTD
CA1->PrefrontalCortex	-0.7000	LTD
PerirhinalCortex->Hippocampus	-0.7000	LTD
Amygdala->PrefrontalCortex	-0.7000	LTD
Amygdala->Thalamus	-0.7000	LTD

EMERGENT THOUGHT LOG (≥75% CONFIDENCE MULTI-REGION CO-ACTIVATION)

Thoughts require ≥3 regions simultaneously co-active above threshold. Silence periods are valid scientific data.

T3240 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 79% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T3210 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 78% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T3180 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 77% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T3150 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 76% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T3120 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 76% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T3090 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 75% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T3060 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 74% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T3030 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 73% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T3000 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 72% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T2970 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 71% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T2940 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 70% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T2910 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 69% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T2880 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 68% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

T2850 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 67% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T2820 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 67% threshold. Brainstem foraging sequence initiated. Metabolic need (14Hz hypothalamic signal) overrides goal-directed behavior."

T2790 · Homeostatic · 80% confidence · Hypothalamus, Brainstem

"Hypothalamic hunger drive at 66% threshold. Brainstem foraging sequence initiated. Metabolic need (15Hz hypothalamic signal) overrides goal-directed behavior."

CORRELATION ANALYSIS (PEARSON R · NOT TRANSFER ENTROPY)

Pearson r (lag-1) = symmetric linear association. Transfer entropy = directional information-theoretic measure. These are NOT interchangeable.

REGION PAIR	PEARSON R	VALIDITY
Thalamus->MotorCortex	0.0004	OK
Thalamus->SensoryCortex	0.0003	OK
MotorCortex->Cerebellum	0.0000	OK
PrefrontalCortex->Thalamus	0.0000	OK
PrefrontalCortex->BasalGanglia	0.0000	OK

EXECUTIVE SUMMARY

- [1] Arousal Pattern: The avatar maintained a low average arousal of 29% across 4880 ticks (peak: 45%). Low arousal suggests dominance of inhibitory tone, likely reflecting resting-state default mode network activity.
- [2] Emotional Trajectory: The avatar's affective state was relatively stable over the session. Valence stability suggests balanced excitatory/inhibitory regulation, a hallmark of healthy homeostatic control.
- [3] Synaptic Plasticity (STDP): The strongest synaptic potentiation occurred at Hippocampus->PrefrontalCortex, PrefrontalCortex->Hippocampus, Thalamus->PrefrontalCortex. These connections strengthened because both pre- and post-synaptic regions fired in correlated temporal proximity – the core mechanism of Hebbian learning. This implies the avatar's neural architecture is self-organizing around the most frequently co-activated circuits.
- [4] Behavioral Patterns: The dominant behavioral state was "RESTING" (resting: 6684). Resting state predominance suggests the simulation spent significant time in low-drive default mode, common during low-complexity or low-arousal conditions.
- [5] Cardiac/ANS Summary: Average heart rate was 60 BPM. Low resting heart rate reflects strong parasympathetic tone with dominant hypothalamic and serotonergic modulation – the autonomic baseline of a calm, regulated state.

BEHAVIORAL EVENTS

TICK	TYPE	DESCRIPTION
T4887	SURGE	Hippocampal theta burst -> memory encoding/retrieval
T4887	SURGE	PFC activation -> executive override of limbic response
T4887	CASCADE	STRATEGY_SHIFT: repeated failure on 'investigate' (3x) -> depressed fronto-striatal pathway -> promoting 'retreat'
T4886	DROP	dACC quiescence -> reduced cognitive conflict monitoring
T4886	DROP	NAC silencing -> reward expectancy drop
T4886	DROP	Insula silencing -> visceral awareness reduced
T4886	DROP	Cerebellar silencing -> motor coordination drop
T4886	DROP	Thalamic suppression -> sensory filtering engaged
T4886	DROP	Sensory cortex drop -> attenuated perception
T4886	DROP	Motor cortex silencing -> movement cessation
T4886	DROP	PFC deactivation -> reduced cognitive control
T4886	CASCADE	STRATEGY_SHIFT: repeated failure on 'pause' (3x) -> depressed fronto-striatal pathway -> promoting 'investigate'
T4885	SURGE	Dorsal ACC surge -> cognitive control, conflict signaling

TICK	TYPE	DESCRIPTION
T4885	SURGE	NAc dopamine surge -> reward prediction signal
T4885	SURGE	Insula activation -> interoceptive salience signal
T4885	SURGE	Cerebellar activation -> precision motor control
T4885	SURGE	Hippocampal theta burst -> memory encoding/retrieval
T4885	SURGE	Thalamic relay surge -> cortical gating cascade
T4885	SURGE	Sensory cortex activation -> multimodal input processing
T4885	SURGE	Motor cortex surge -> voluntary movement initiation